Reverse Mode & Backprop

A scalar loss gradient in one reverse sweep

Prof. Nipun Batra

Mathematical Foundations for AI · IIT Gandhinagar

What `loss.backward()` computes

A common operation in gradient-based training

```
loss = f(parameters, data)    # forward pass: ONE number comes out
loss.backward()               # ← this line. What does it DO?
```

Many modern models are trained by repeating a forward evaluation of a scalar loss and a reverse-mode gradient computation.

After one reverse sweep, $\partial \text{loss}/\partial \theta$ is available for every parameter that contributed to the loss. The runtime is a small constant multiple of the forward evaluation, not one pass per parameter.

We will compute the reverse pass by hand, implement a scalar autodiff engine in about 35 lines of Python, and compare it with PyTorch.

Reverse mode for a scalar output

Reverse mode computes the gradient of one output with respect to a million inputs in one backward pass.

This input/output asymmetry makes gradients of scalar losses computationally practical even when the parameter vector is large.

Three questions, in order:

1. What structure hides inside an expression? → **computation graphs**
2. What flows backward through it? → **adjoints** (the chain rule, organized)
3. Who does the bookkeeping? → **you** (micrograd), then **PyTorch**

The cost left open in L10

Method	**Assessment from L10**	**Cost for ∇f, $f : \mathbb{R}^n \to \mathbb{R}$**
symbolic	exact, but expressions explode	—
finite differences	truncation vs rounding — both lose	n evals, **approximate**
forward-mode AD	no finite-difference approximation; carry $(\text{value}, \text{derivative})$ pairs	n **passes** — one per input

Forward mode removed the finite-difference approximation, but a million input directions still require a million passes to assemble the full gradient.

The cost left open in L10

Reverse mode seeds the scalar output and propagates adjoints backward, producing every input partial derivative in one reverse sweep.

Learning outcomes

By the end of this lecture you will be able to:

1. Draw the **computation graph** of any small expression — nodes, edges, DAG.
2. Run a **forward pass** and a complete **backward pass** by hand, node by node.
3. Define the **adjoint** $\overline{v} = \partial f / \partial v$ and apply "arriving × local, add at branches".
4. Explain the **cost asymmetry**: one pass per input (forward) vs one pass per output (reverse).
5. Implement a working scalar autograd engine — **micrograd** — and verify it.
6. Reproduce everything in PyTorch: `requires_grad`, `.backward()`, `.grad`, `zero_grad()`.

Computation graphs

You already compute in graphs

Evaluate $f(x, y) = (x + y) \cdot x^2$ at $(3, 1)$. Watch what your own hand does:

$$a = x + y = 4, \qquad b = x^2 = 9, \qquad f = a \cdot b = 36$$

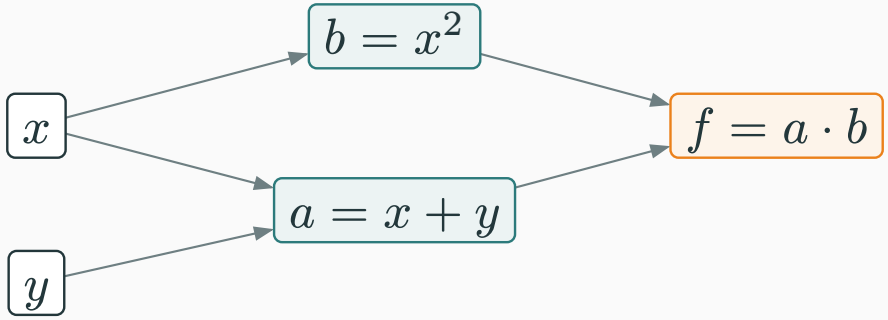
Three small steps, each a **primitive** — a single operation whose derivative we know cold ($+$, $\times$, $(\cdot)^2$, and later $\exp$, $\tanh$, ...).

Each step **names** an intermediate value, and each step **uses** earlier values.

An expression is a recipe: primitive steps, wired by "who uses whom". Draw the wiring and you have the computation graph.

The expression, drawn

V VISUAL



- **Leaves** (left): the inputs x, y. **Root** (right): the output f.
- Every arrow says “this value feeds that operation”.

Look at x: it feeds **two** nodes — it is written once but **used twice**. So the picture is a **DAG** (directed acyclic graph), not a tree. Remember this; it is where the one subtle rule of the day will live.

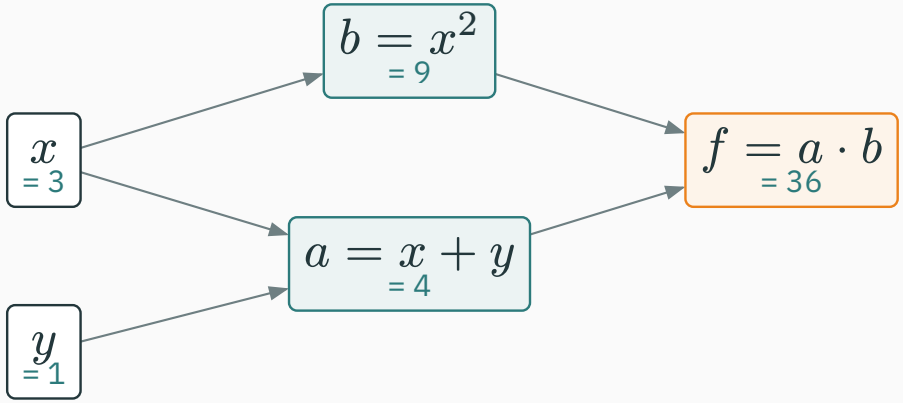
Why graphs? Because computers already build them

- When Python evaluates `(x + y) * x**2`, it **must** execute primitives in some order — the graph is just that execution, remembered.
- PyTorch, JAX, TensorFlow all do exactly this: as your code runs, every operation quietly **records** itself: result, operands, and which primitive fired.
- The recording is called the **tape** (or “autograd graph”). Recording costs almost nothing — the work was being done anyway.

No new math yet: a computation graph is bookkeeping for work the machine already did.

Forward pass: values flow along the arrows

V VISUAL



Node	Rule	Value
$a = x + y$	$3 + 1$	4
$b = x^2$	3^2	9
$f = a \cdot b$	$4 \cdot 9$	36

This is the **forward pass**: leaves are given, every other node fires once its inputs are ready.

Keep every number in sight — the backward pass is about to **reuse all of them**.

Adjoints: the chain rule on a graph

L8′s chain rule, re-read as a graph rule

L8 gave the multivariate chain rule. For our $f(a(x, y), b(x))$:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial a}\frac{\partial a}{\partial x} + \frac{\partial f}{\partial b}\frac{\partial b}{\partial x}$$

Read it **on the graph**:

> **Multiply local slopes along a path. Add over all paths from the variable to f.**

A deep graph can hold **exponentially many** paths — summing them one by one would be hopeless. Today's algorithm computes each node's "all paths downstream of me" summary **once**, and reuses it. Hold that thought: it is **memoization**.

The adjoint: one number per node

For every node v in the graph, define its **adjoint**

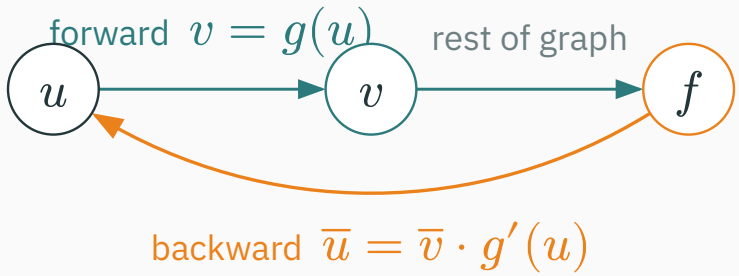
$$\overline{v} := \frac{\partial f}{\partial v} \quad = \quad \text{"if } v \text{ wiggles by } \varepsilon, \quad f \text{ wiggles by } \overline{v} \cdot \varepsilon\text{"}$$

- The adjoint already accounts for **everything between v and the output** — all paths, summed.
- At the output itself: $\overline{f} = \partial f / \partial f = 1$. Free seed, no work.
- The two adjoints we actually want: $\overline{x}$ and $\overline{y}$ — together they **are** ∇f.

Notation: the bar $\overline{v}$ is standard in the autodiff literature; PyTorch stores exactly this number in `v.grad`.

Each node's job: arriving × local

V VISUAL



The node $v = g(u)$ never sees the whole graph. It receives $\overline{v}$ from above, multiplies by its **own local slope**, and passes the product down:

$$\overline{u} = \overline{v} \cdot \underbrace{g'(u)}_{\text{local, known cold}}$$

derivative sent back = derivative arriving × local slope

The three local rules we need today

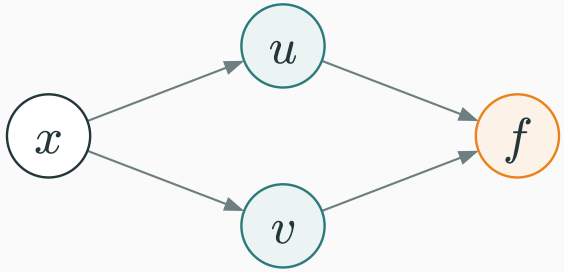
Node	**Local slopes**	**Backward behaviour**
$v = u + w$	$\partial v/\partial u = 1, \quad \partial v/\partial w = 1$	$+$ **copies** $\overline{v}$ to both inputs
$v = u \cdot w$	$\partial v/\partial u = w, \quad \partial v/\partial w = u$	$\times$ **sends each input the other input**: $\overline{u} = \overline{v} \cdot w$
$v = u^2$	$\partial v/\partial u = 2u$	square sends $\overline{v} \cdot 2u$

Every primitive an engine supports ships with its one-line rule like these — $\exp$ sends $\overline{v} \cdot e^u$, $\tanh$ sends $\overline{v}(1 - \tanh^2 u)$, ...

An autodiff engine is: this table, plus bookkeeping.

Where paths meet, contributions add

V VISUAL



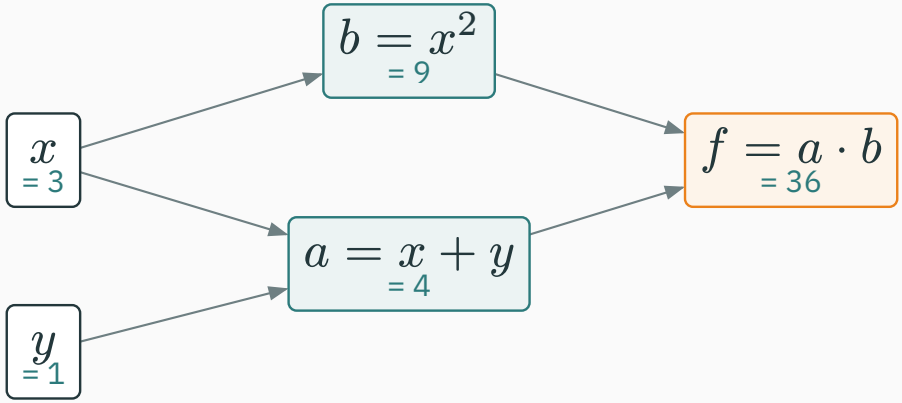
If x feeds two nodes, a nudge Δx travels down **both** routes to f, and the two effects superpose (first-order Taylor — L7):

$$\Delta f \approx \underbrace{\overline{u}\frac{\partial u}{\partial x}}_{\text{via } u} \cdot \Delta x + \underbrace{\overline{v}\frac{\partial v}{\partial x}}_{\text{via } v} \cdot \Delta x \quad \Rightarrow \quad \overline{x} = \text{sum of both contributions}$$

The one subtle rule of the day: at a branch point, backward contributions **accumulate** (+=), never overwrite. Our graph's x is exactly such a point — watch it in the derivation.

⭐ The backward pass, by hand

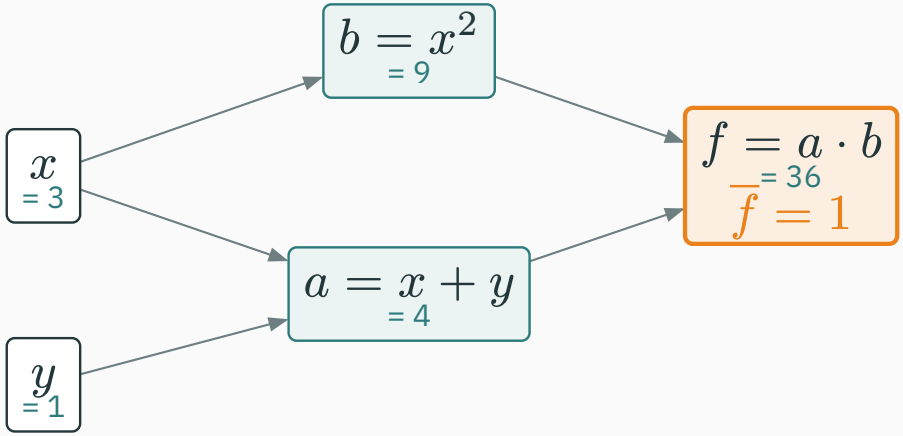
The plan of attack



1. **Seed** the output: $\overline{f} = 1$.
2. Visit nodes from the output back toward the inputs — each node only after **everything it feeds** is done. Here: f, then b, then a.
3. At each node: send **arriving × local** down every incoming edge; where edges meet, **add**.

Step 0 · seed the output

D DERIVATION

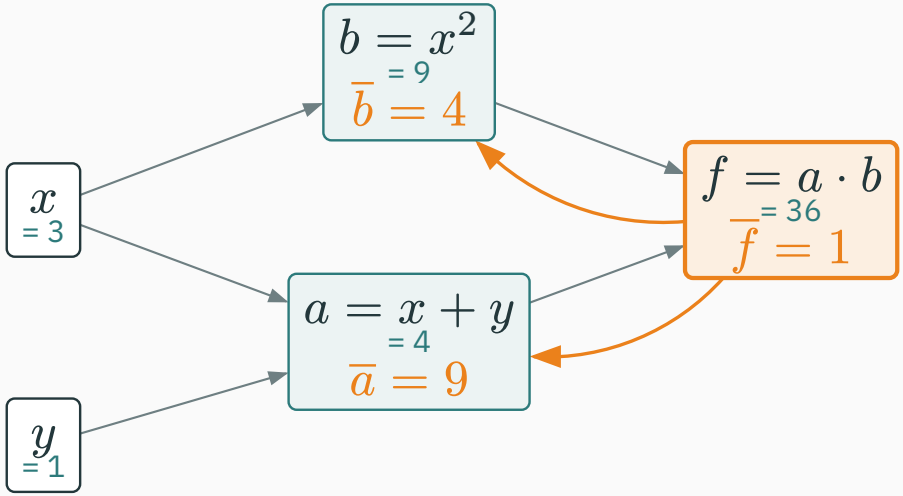


$$\overline{f} = \frac{\partial f}{\partial f} = 1 \qquad \text{(a value can't help but move one-for-one with itself)}$$

Every other adjoint will now follow from the local-rules table — no calculus, only arithmetic.

Step 1 · the × node fires

D DERIVATION



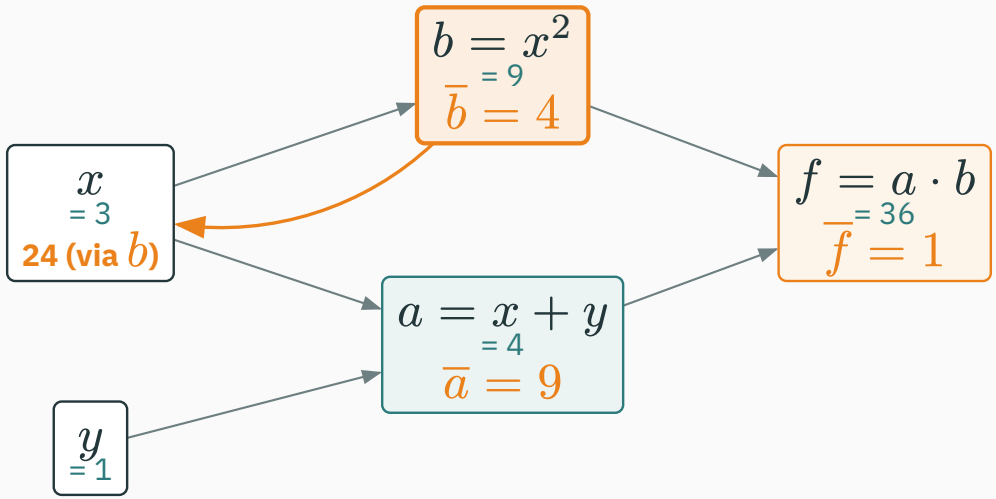
$f = a \cdot b$: **send each input the other input.**

$$\overline{a} = \overline{f} \cdot b = 1 \cdot 9 = 9 \qquad \overline{b} = \overline{f} \cdot a = 1 \cdot 4 = 4$$

With $\overline{f} = 1$, each factor's adjoint **is the other factor's value** — read both off the forward pass, zero new work.

Step 2 · the square node fires

D DERIVATION



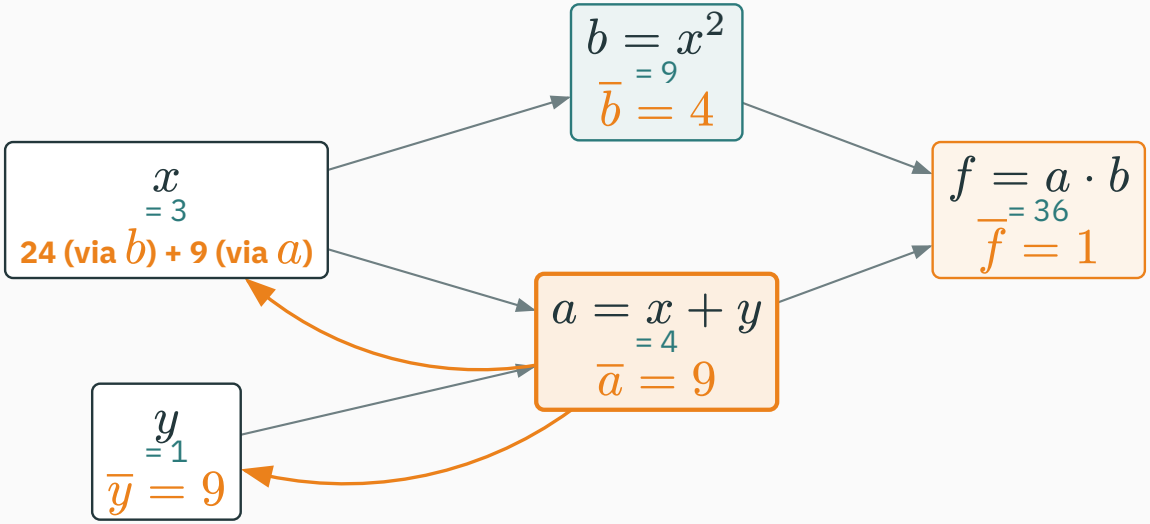
$b = x^2$: local slope $2x = 6$. Arriving × local:

$\overline{b} \cdot 2x = 4 \cdot 6 = 24$ flows down the $b \leftarrow x$ edge

Do **not** write $\overline{x} = 24$ yet! A second path into x (via a) has not reported. Park the contribution and wait.

Step 3 · the + node fires

D DERIVATION



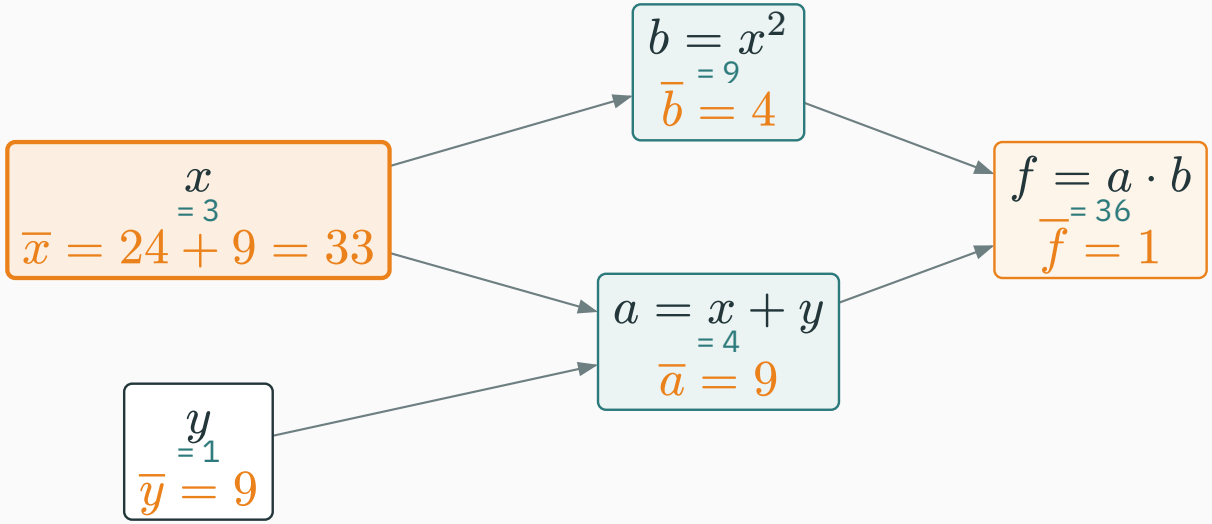
$a = x + y$: **copy the arriving adjoint to both inputs** (local slopes are 1).

to x : $\overline{a} \cdot 1 = 9$ to y : $\overline{a} \cdot 1 = 9$

y has only this one path, so it is done: $\overline{y} = 9$.

Step 4 · accumulate at the branch

D DERIVATION



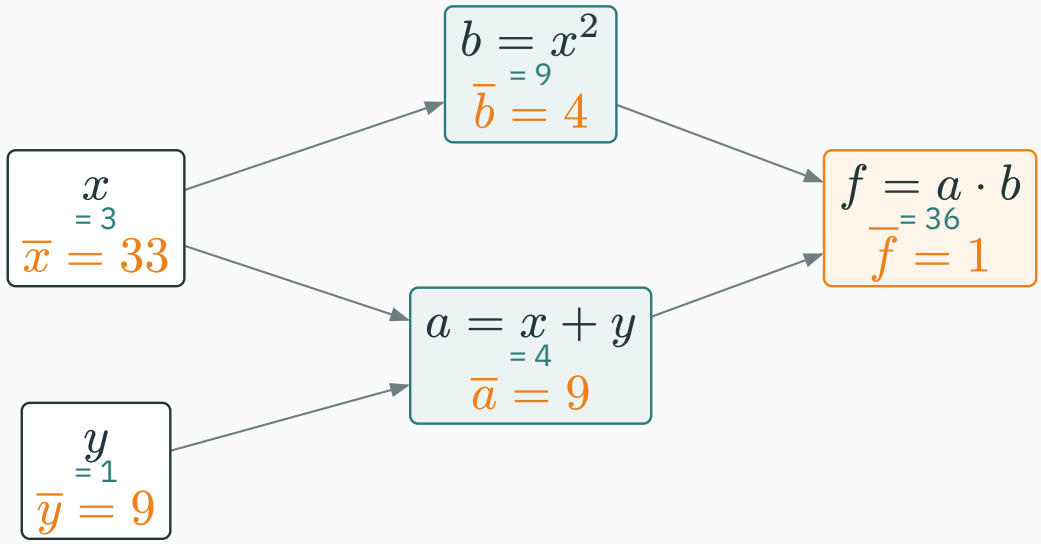
Both of x's paths have reported — the branch rule says **add**:

$$\overline{x} = \underbrace{24}_{\text{via } b} + \underbrace{9}_{\text{via } a} = 33$$

$\nabla f(3,1) = (\overline{x}, \overline{y}) = (33, 9)$ **— every input's derivative, one sweep.**

The finished graph — photograph this

V VISUAL



teal: forward values, flowing right · orange: adjoints, flowing left — same graph, two sweeps

The procedure you just executed has a name: **backpropagation**.

Tutorial 5 re-runs this exact graph with fresh numbers until it is reflex; **Quiz 2** assumes it is.

Does calculus agree?

Expand and differentiate the old way: $f = (x + y)x^2 = x^3 + x^2y$.

$$\frac{\partial f}{\partial x} = 3x^2 + 2xy = 27 + 6 = 33 \quad \checkmark \qquad \frac{\partial f}{\partial y} = x^2 = 9 \quad \checkmark$$

Same numbers. But look **how** each method got there:

- Calculus needed the **global formula** for f, expanded and re-derived per input.
- Backprop only ever used **local** facts — a table of primitive rules and the forward values. Locality is what a computer can automate.

This slide ran a backward pass while compiling

These slides ship their own reverse-mode engine (chalkdust `autodiff`, written in Typst). This slide **runs it** on today's expression — the numbers below are computed live, not typed:

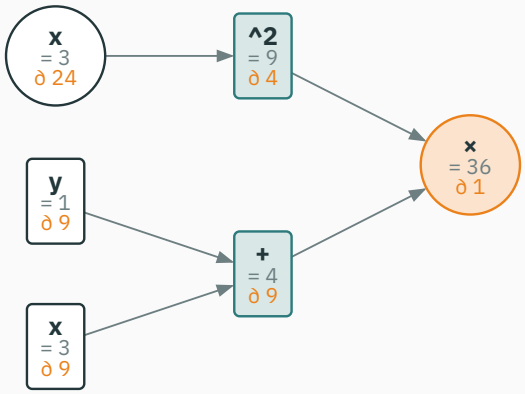
```
#let f5 = ad.expr("(x + y) * x^2", ("x", "y"))
#ad.value(f5, (3, 1))   // forward pass
#ad.grad(f5, (3, 1))    // one backward pass → (x̄, ȳ)
```

engine says: $f = 36$, $\nabla f = (33, 9)$ — the slide agrees with your hand.

Hand, calculus, Typst engine: three routes, one answer. Two more coming (micrograd, PyTorch).

The autodiff tape as a computation graph

V VISUAL



auto-drawn from the recorded tape: each node shows op, value (“=”), adjoint (“∂”)

The tape records each **use** of x as its own entry — so the two x leaves carry their per-path contributions 24 and 9 separately. The variable x collects both: $24 + 9 = 33$. The accumulation rule, made visible.

Checkpoint: run the graph at a new point

Q QUESTION

Same graph, new point: $x = 2, y = 5$ **— so** $a = 7$**,** $b = 4$**,** $f = 28$**. What is** $\overline{y}$**?**

A. 7

B. 4

C. 28

D. 32

Answer: one path for y

A ANSWER

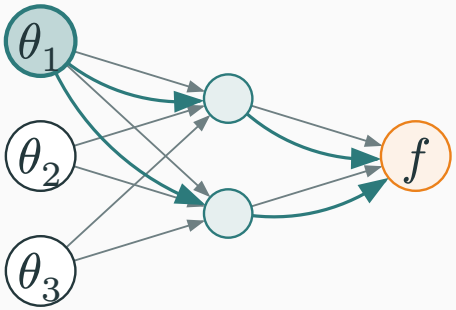
Correct: B — $\overline{y} = 4$

Why: y reaches f through one path: the $+$ node **copies** $\overline{a}$, and the $\times$ node made $\overline{a} = \overline{f} \cdot b = 4$. Sanity check by calculus: $\partial f / \partial y = x^2 = 4$. (Option D is $\overline{x} = 3x^2 + 2xy = 32$ — the **other** input's adjoint.)

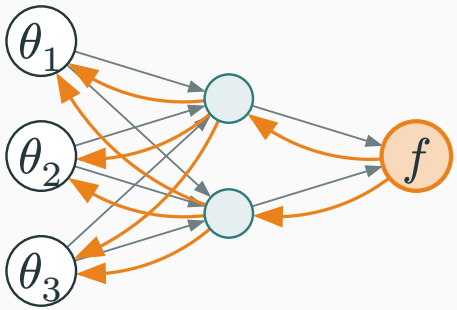
When reverse mode is cheaper

Two sweep directions, two prices

V VISUAL



forward mode (L10): seed **one input**, push tangents up — yields $\partial f / \partial \theta_1$ only



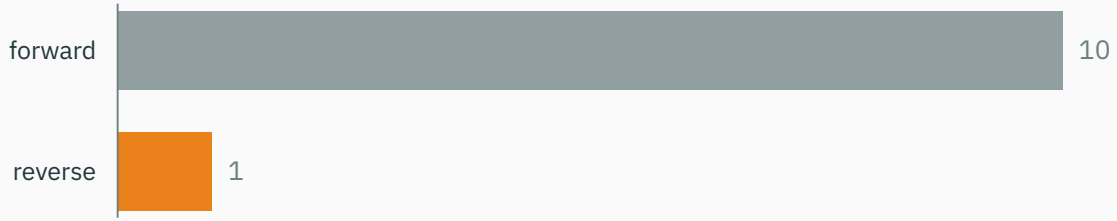
reverse mode (today): seed **the output**, push adjoints down — yields **all** $\partial f / \partial \theta_i$

Forward: one pass per input. **Reverse: one pass per output.**

Pass counts for a full gradient

V VISUAL

Passes needed to assemble the full gradient of $f : \mathbb{R}^{10} \to \mathbb{R}$:



And the gap **scales**:

Function	**Inputs n**	**Forward-mode passes**	**Reverse passes**
today's $f(x, y)$	2	2	1
small image model's loss	10^7	10^7	1
GPT-3's loss	1.75×10^{11}	1.75×10^{11}	**1**

Pass counts for a full gradient

V VISUAL

At 60 ms per directional pass, the forward-mode estimate is about **330 years** to assemble this full gradient. Reverse mode needs one sweep, typically comparable to a small number of forward evaluations.

Scalar losses favor reverse mode

Why does reverse mode fit machine learning so perfectly?

- Training tunes n = **millions to billions** of inputs — the parameters θ.
- But it scores them with m = **one** output — the loss $\mathcal{L}(\theta)$ (L14 builds it: a log-likelihood).
- $\mathbb{R}^n \to \mathbb{R}$, with large n and $m = 1$, is the regime in which reverse mode has the favorable pass count.

For a scalar loss with many parameters, one reverse sweep computes the full gradient; a full forward-mode gradient needs one seed direction per input.

For $g : \mathbb{R} \to \mathbb{R}^{10^6}$ (one input, many outputs), forward mode needs one pass whereas reverse mode needs one pass per output coordinate.

The price: memory (here is the memoization)

Reverse mode is not free. Look what the backward steps **consumed**:

- the $\times$ rule needed the stored values $a = 4$, $b = 9$
- the square rule needed the stored $x = 3$ (for $2x$)

So the forward pass must **keep every intermediate value alive** until backward is done — that stored trail is the tape.

Store each node's value once, reuse it for every path through that node: backprop is the chain rule with memoization.

The trade: a single forward-mode directional derivative can stream without retaining a reverse tape; reverse mode stores intermediate values, using

The price: memory (here is the memoization)

memory proportional to the recorded graph. This is one major reason training needs more memory than inference.

The cheap-gradient principle

How much **time** does the backward sweep add? Each node fires once, doing constant work per edge — the same order of work as the forward pass itself.

Computing ∇f costs a small constant multiple (≈ 2–$3\times$) of computing f — independent of n. True for $n = 2$ and for $n = 1.75 \times 10^{11}$.

This is the **cheap gradient principle** (Baur–Strassen; Griewank): for suitable computational graphs, a scalar-output gradient costs only a constant factor more arithmetic than evaluating the function. Reverse mode pays for that time efficiency with storage for the tape.

Checkpoint: pick the mode

Q QUESTION

$f : \mathbb{R}^{10^6} \to \mathbb{R}$ (a loss). $g : \mathbb{R} \to \mathbb{R}^{10^6}$ (one input, a million outputs). Which mode computes all required derivatives cheapest?

A. forward for both

B. reverse for f, forward for g

C. forward for f, reverse for g

D. reverse for both

Answer: match the pass to the thin side

A ANSWER

Correct: B — reverse for f, forward for g

Why: Passes = one per input (forward) or one per output (reverse). f has one output → 1 reverse pass beats 10^6 forward passes. g has one input → 1 forward pass beats 10^6 reverse passes. Always seed the **thin** side.

Micrograd: build the engine

What a number must remember

To automate the hand-run, each value needs to carry exactly what **you** used at the desk:

Field	What it held in the hand-run
`data`	the forward value (teal numbers: $3, 1, 4, 9, 36$)
`grad`	the adjoint slot, starting at 0 (orange numbers, `+=` into it)
`_prev`	which nodes fed me (the arrows)
`_backward`	my one line from the local-rules table, frozen as a function

We now write this class in three stages — it is Karpathy's **micrograd**, the engine his assigned video builds. Same fields, same names.

Stage 1 · a `Value` that remembers its history

```
class Value:
    """A scalar that knows how it was computed."""
    def __init__(self, data, _children=()):
        self.data = data                # forward value
        self.grad = 0.0                 # adjoint: d(output)/d(self)
        self._backward = lambda: None   # local rule (leaves: nothing)
        self._prev = set(_children)     # the nodes that made me
```

- A plain number, plus the graph bookkeeping. `grad = 0.0` so contributions can `+=` in.
- Leaves (x, y) are just `Value(3.0)`, `Value(1.0)` — no children, nothing to send back.

Stage 2a · addition, carrying its backward rule

```
    def __add__(self, other):                   # self + other
        out = Value(self.data + other.data, (self, other))
        def _backward():
            self.grad  += out.grad              # + copies the
            other.grad += out.grad              #   arriving adjoint
        out._backward = _backward
        return out
```

- The forward line computes the value **and wires the graph** (`(self, other)` become `out._prev`).
- The closure is the rules-table row for $+$, waiting: “when my turn comes, copy `out.grad` down”. Note the `+=` — branch accumulation, built in.

Stage 2b · multiplication: send the other input

```
    def __mul__(self, other):                    # self * other
        out = Value(self.data * other.data, (self, other))
        def _backward():
            self.grad  += other.data * out.grad   # × sends each
            other.grad += self.data  * out.grad   #   the OTHER input
        out._backward = _backward
        return out
```

- `other.data * out.grad` is precisely **arriving × local** — the local slope of $u \cdot w$ w.r.t. u **is** w.

Each operator owns one closure = one row of the rules table. The engine never needs a global formula.

In what order may nodes fire?

A node may push its `grad` down **only when that `grad` is complete** — after every node it feeds has already fired.

- Fire a before f? Then a pushes `a.grad = 0` — silence — and x, y end up wrong forever.
- Safe order for our graph: f, then b, then a (leaves receive; they never push).
- The general recipe: list nodes so every node comes **after** all nodes it feeds — a **topological order** of the DAG, walked in reverse.

Getting this order is a classic graph traversal: depth-first, children before self ("post-order"). Ten lines, next slide.

Stage 3 · topological sort, then one sweep

```
    def backward(self):
        topo, seen = [], set()
        def build(v):                       # depth-first:
            if v not in seen:
                seen.add(v)
                for c in v._prev:           #   children first,
                    build(c)
                topo.append(v)              #   then me
        build(self)
        self.grad = 1.0                     # seed: df/df = 1
        for v in reversed(topo):            # output → inputs
            v._backward()
```

Seed, sweep, done — steps 0–4 of the hand derivation, mechanized in twelve lines.

Run it on the lecture's graph

```
x, y = Value(3.0), Value(1.0)
f = (x + y) * x * x        # the same 5-node graph
f.backward()

print(f.data)              # 36.0
print(x.grad, y.grad)      # 33.0  9.0
```

36, 33, 9 — **the hand-run, reproduced by 35 lines of Python you can hold in your head.**

We wrote `x * x` because our `Value` has no power op yet. Watch the mul closure: **both** branches are `x`, so it fires `x.grad += x.data * out.grad` twice — the $2x$ rule emerges from accumulation, unasked.

35 lines, and it is real

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Walk a computation graph in the browser: symbolic vs finite-difference vs autodiff on one screen, forward values then a step-by-step backward sweep. Rebuild today's $f = (x + y) \cdot x^2$ in it and watch 33 and 9 appear.

Assignment 1 (micrograd) goes out this week. You take today's 35 lines and grow them Karpathy's way: `tanh`, `exp`, `__pow__`, `__neg__`, broadcasting scalars — then use **your own engine** to fit a small model. The assigned video walks the full 94-line original; watch it before you start.

PyTorch: the industrial engine

The same numbers, a fifth and final time

```python
import torch

x = torch.tensor(3.0, requires_grad=True)
y = torch.tensor(1.0, requires_grad=True)

f = (x + y) * x**2       # forward — PyTorch records the tape
f.backward()             # reverse sweep

print(f.item())          # 36.0
print(x.grad, y.grad)    # tensor(33.)  tensor(9.)
```

Hand · calculus · Typst engine · micrograd · PyTorch: five routes, one gradient $(33, 9)$.

What PyTorch adds to the scalar implementation

Same algorithm as your 35 lines. Different engineering:

- Nodes hold **tensors**, not scalars — one graph node per matrix op, not per number.
- Each primitive (`matmul`, `exp`, `softmax`, …) ships a hand-tuned backward rule, running on GPU.
- The tape is rebuilt **on every forward run** (“define-by-run”) — Python `if`s and loops just work.
- `requires_grad=True` marks which leaves deserve adjoints; everything else is constants.

That is genuinely all. If you can read your micrograd, you can read `torch.autograd`’s documentation without fear.

Careful: gradients accumulate

`.grad` is an accumulator — `backward()` always `+=`s into it:

```
f = (x + y) * x**2
f.backward()
print(x.grad)            # tensor(33.)

f = (x + y) * x**2       # build the graph again…
f.backward()             # …sweep again
print(x.grad)            # tensor(66.)  ← 33 + 33. Not 33!
```

Why design it this way? The engine cannot know your intent — the same `+=` that merged x's two paths **inside** one graph also merges gradients **across** calls (useful for micro-batching). Clearing is your job.

What `zero_grad()` is for

```
for step in range(1000):
    optimizer.zero_grad()   # 1 · flush stale adjoints
    loss = compute_loss()   # 2 · forward (tape records)
    loss.backward()         # 3 · reverse sweep → .grad
    optimizer.step()        # 4 · nudge parameters (L16!)
```

If line 1 is omitted, gradients from previous iterations accumulate silently. Clear them unless accumulation across batches is intentional.

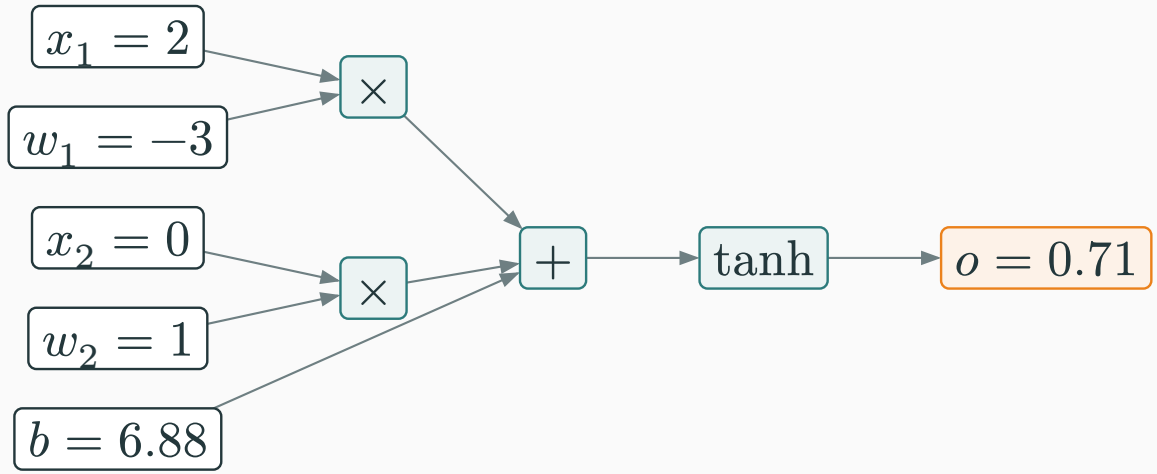
Lines 3–4 are the bridge to Module 4: L16 takes the gradients you can now compute and asks **where to step**.

Example: differentiate one neuron

V VISUAL

One last graph — small algebra, but shaped like the real thing: inputs weighted, summed, squashed.

$o = \tanh(w_1 x_1 + w_2 x_2 + b)$



This unit is an (artificial) **neuron** — the atom of every network. To the engine it is nothing special: five leaves, two $\times$, one $+$, one $\tanh$. Backprop needs no new ideas.

Differentiate one neuron with the same engine

```
x1, x2 = torch.tensor(2.0), torch.tensor(0.0)
w1 = torch.tensor(-3.0, requires_grad=True)
w2 = torch.tensor( 1.0, requires_grad=True)
b  = torch.tensor( 6.88137, requires_grad=True)

o = torch.tanh(w1*x1 + w2*x2 + b)   # o = 0.7071
o.backward()
print(w1.grad, w2.grad, b.grad)     # 1.0   0.0   0.5
```

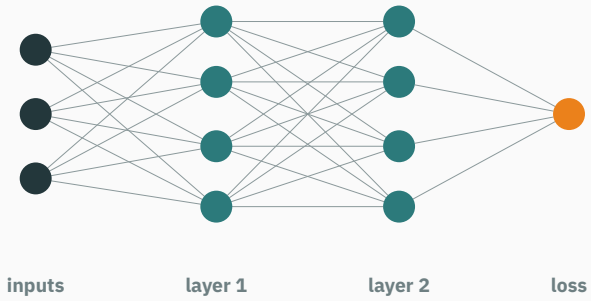
Our Typst engine, live, agrees: $\overline{w}_1 = 1$, $\overline{w}_2 = 0$, $\overline{b} = 0.5$.

Read $\overline{w}_2 = 0$ off the mul rule: $\times$ **sends the other input**, and w_2's partner is $x_2 = 0$. A weight attached to a silent input gets no gradient — no data, no learning signal.

What `backward()` does, itemized

Every step of `loss.backward()` is now one you have done yourself:

1. The forward pass already **recorded the tape** (your `_prev` wiring).
2. Topological sort of the tape (your twelve-line `build`).
3. Seed $\overline{\text{loss}} = 1$; sweep once: **arriving × local**, `+=` at branches.
4. Adjoints land in `.grad` — millions at a time.



What `backward()` does, itemized

The same reverse-sweep procedure applies to larger graphs; work scales with the graph rather than with one separate evaluation per parameter.

Summary & what’s next

Lecture 11 — summary

- **Graphs**: an expression is a DAG of primitives; the forward pass fills every value.
- **Adjoints**: $\overline{v} = \partial f / \partial v$; seed $\overline{f} = 1$; each node sends **arriving × local**; branches `+=`.
- **The graph**: $f = (x + y)x^2$ at $(3, 1)$: $f = 36$, $\nabla f = (33, 9)$ — five methods, one answer.
- **Asymmetry**: one pass per input (forward) vs one pass per **output** (reverse); a loss has one output.
- **The price**: the tape stores forward values — extra memory in exchange for avoiding one pass per input.
- **Engines**: `data / grad / _prev / _backward` + topo sort + one sweep; in PyTorch `requires_grad` → `.backward()` → `.grad`, with `zero_grad()` every step.

Read before L12 — MML §5.6. **Assigned:** Karpathy, “spelled-out intro to backpropagation” (builds micrograd; watch before starting A1). ★★★ JAX autodiff cookbook. **This week:** T5 drills today’s graph by hand · **Quiz 2** (L7–L11) · **A1 (micrograd)** goes out.

Practice problems

Try on paper; verify with the T5 notebook (or your own micrograd).

1. Draw the graph of $g(x, y) = (x - y) \cdot (x + y)$; run forward and backward at $(3, 1)$. Check both adjoints against $g = x^2 - y^2$.
2. Re-run today's graph at $x = 1, y = 2$. Before computing: which of $\overline{x}, \overline{y}$ changes, and why?
3. Give `Value` a `__sub__`. What are its two local slopes, and its two closure lines?
4. In `backward()`, delete `reversed(...)` and trace our graph. What does `x.grad` end as, and which silent push caused it?
5. Predict PyTorch: build and `backward()` the same loss twice with no `zero_grad()` between. What is `w.grad`? Which single line fixes it?
6. For $h : \mathbb{R}^3 \to \mathbb{R}^2$, count passes to get the full Jacobian by each mode. Name a function shape where **forward** mode is the right tool.

⭐⭐⭐ JVPs, VJPs, and Hessians without Hessians

★ OPTIONAL

Both modes are matrix-free products with the Jacobian J of $f : \mathbb{R}^n \to \mathbb{R}^m$ (L9):

Sweep	**Computes**	**Reading**
forward, seeded with $\boldsymbol{v} \in \mathbb{R}^n$	$J\boldsymbol{v}$ — a **JVP**	one **column** mix: sensitivity along direction $\boldsymbol{v}$
reverse, seeded with $\boldsymbol{u} \in \mathbb{R}^m$	$\boldsymbol{u}^\top J$ — a **VJP**	one **row** mix: ∇f is the VJP with $u = 1$

Compose them and curvature comes free: for scalar f, the **Hessian-vector product**

$$H\boldsymbol{v} = \nabla(\nabla f(\theta)^\top \boldsymbol{v}) \quad = \quad \text{forward mode pushed through the reverse-mode program}$$

costs a few function evaluations and **never materializes** the $n \times n$ Hessian — the trick behind Newton-flavoured optimizers (L18) at scales where n^2 storage is fiction. JAX exposes all three as `jvp`, `vjp`, `grad`; see the autodiff cookbook.

Backprop is just the chain rule with memoization.

Next: **Module 3 — Continuous Distributions** — we can now differentiate anything; next we need something worth differentiating: likelihoods.